

RLS-Enabled Regional Sales & Profit Analytics

Power BI Project-Based Assignment: Final Report

Dataset: Indian Superstore (Orders and Zone sheets), January 2023 to December 2025

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1. Background and Objective

A retail organisation sells technology and office supplies across regions in India. Management wants to monitor sales, profit, customer behaviour and shipping performance, and to measure region-wise manager performance securely. This project builds an interactive Power BI Sales & Profit dashboard on the Superstore dataset.

Objectives:

- Build an interactive Sales & Profit dashboard.
- Implement Row-Level Security (RLS) so each Regional Manager sees only their own region.
- Create drill-down, drill-through, custom tooltips and bookmarks.
- Provide insights across Category, Region, Customer and Product levels.
- Enable data-driven business decisions through visual analytics.

Headline results:

- Total Sales: ₹104.93 Cr
- Total Profit: ₹12.18 Cr
- Profit Margin: 11.6%
- Total Orders: 51,236

2. Phase 1: Data Import and Cleaning (Power Query)

The workbook has two sheets: Orders (51,290 rows, 24 columns) and Zone (5 rows mapping each Region to its Region Manager). After cleaning, Orders has 51,284 rows and 23 columns.

Power Query transformation notes

1. Loaded the Excel workbook (Orders and Zone sheets).
2. Promoted the first row to headers on Zone (headers were read as Column1, Column2).
3. Set data types (Date, Fixed decimal, Whole number, Text, Decimal). Dates carried a 12:00 AM time part, removed by converting to Date.
4. Removed Postal Code (100% blank) and Market (single value).
5. Filtered Order Date on or after 01-01-2023. Six rows dated 2012 were invalid; rows went from 51,290 to 51,284.
6. Removed duplicates on Row ID, the unique key. No duplicates were found.

7. Added the custom column Delivery Days = Duration.Days([Ship Date] - [Order Date]) as a Whole Number.
8. Trimmed and cleaned all text columns to remove stray spaces and hidden characters.
9. Trimmed and cleaned Region and Region Manager in Zone so the relationship and RLS match exactly.
10. Close and Apply.

Data quality findings

- No duplicate rows, no negative or zero Sales or Quantity, Discount between 0% and 20%, and Ship Date never before Order Date.
- Order ID repeats across order lines (51,236 distinct orders), so Total Orders uses a distinct count.
- Customer ID is almost unique per row, so customer analysis uses Customer Name (945 customers).
- Product Name has 40 distinct values, so product analysis uses Product Name.

Data dictionary

- Order Date (Date, Orders): date the order was placed
- Ship Date (Date, Orders): date the order was shipped
- Row ID (Whole number, Orders): unique key of each order line
- Order ID (Text, Orders): order number, repeats across lines
- Ship Mode (Text, Orders): Standard, Second Class, First Class, Same Day
- Customer ID and Customer Name (Text, Orders): customer identifiers
- Segment (Text, Orders): Consumer, Corporate, Home Office
- City, State, Country (Text, Orders): delivery geography (56 cities, 22 states)
- Region (Text, Orders and Zone): North, South, East, West, Central; the join key
- Product ID and Product Name (Text, Orders): product identifiers (40 product names)
- Category and Sub-Category (Text, Orders): 3 categories, 17 sub-categories
- Sales, Profit, Shipping Cost (Fixed decimal, Orders): amounts in INR; negative profit is a loss
- Quantity (Whole number, Orders): units on the order line
- Discount (Decimal, Orders): discount rate, 0 to 0.20
- Order Priority (Text, Orders): Low, Medium, High, Critical
- Delivery Days (Whole number, Orders): calculated, Ship Date minus Order Date (5 to 30)
- Region Manager (Text, Zone): manager of the region, used for RLS

3. Phase 2: Data Modeling and DAX

The model is a star schema. Orders is the fact table. Date Table (a calculated table covering 01-Jan-2023 to 31-Dec-2025, 1,096 rows, marked as the date table) and Zone are dimension tables.

[Insert screenshot: Model view / data model diagram]

Relationships (all one to many, single direction, active)

- Date Table[Date] to Orders[Order Date]
- Zone[Region] to Orders[Region]

DAX formula list

- Total Sales = SUM(Orders[Sales])
- Total Profit = SUM(Orders[Profit])
- Profit % = DIVIDE([Total Profit], [Total Sales], 0)
- Total Orders = DISTINCTCOUNT(Orders[Order ID])
- Discount % = AVERAGE(Orders[Discount])
- Avg Delivery Days = AVERAGE(Orders[Delivery Days])

Helper items used by visuals: Discount Band (calculated column grouping Discount into five bands) and Sales Share % (segment contribution).

Verification

- Total Sales: expected ₹1,049,271,545 - matched
- Total Profit: expected ₹121,794,537 - matched
- Profit %: expected 11.61% - matched
- Total Orders: expected 51,236 - matched
- Discount %: expected 10.00% - matched
- Avg Delivery Days: expected 17.52 - matched

4. Phase 3: Dashboard Development

Page 1: Sales Overview

[Insert screenshot: Sales Overview page]

Visuals: KPI cards, Category and Sub-Category chart, Sales by State map, Monthly Sales Trend, Top and Bottom 10 products.

Which category performs the best? Technology: ₹39.38 Cr sales (37.5%), ₹5.51 Cr profit, 14.0% margin. Furniture (₹34.12 Cr, 6.9% margin) and Office Supplies (₹31.43 Cr, 13.7% margin) follow.

Which states drive the highest revenue? Assam (₹5.19 Cr), Madhya Pradesh (₹5.15 Cr), Kerala (₹5.11 Cr), Himachal Pradesh (₹4.95 Cr) and Tamil Nadu (₹4.94 Cr). The top 5 make only 24% of sales, and every state holds 4.1% to 5.0%.

Monthly growth patterns Strong seasonality: July to December is 57% of yearly sales, with November and December at 10.9% each; January and February are the weakest. Yearly sales are flat to slightly lower: ₹35.46 Cr (2023), ₹34.91 Cr (2024), ₹34.56 Cr (2025).

Page 2: Customer Insights

[Insert screenshot: Customer Insights page]

Visuals: segment donut, Top 10 customers by sales, Bottom 10 customers by profit, Region vs Customer Sales, with drill-through to the Customer Profile page.

Who are our top value customers? Sana Bhattacharya (₹9.0 lakh profit), Siddharth Bhattacharya (₹8.2 lakh), Nikhil Ghosh (₹6.4 lakh) and Lokesh Nair (₹5.6 lakh) rank high on both sales and profit. Nikhil Sharma has the highest sales (₹35.9 lakh) but a loss of about ₹1.1 lakh. The top 10 customers make only 2.3% of sales, and 84 of 945 customers are unprofitable.

Segment contribution to sales Consumer 51.5% (₹54.01 Cr), Corporate 30.3% (₹31.74 Cr) and Home Office 18.3% (₹19.17 Cr). Margins are similar (11.5% to 12.0%).

Page 3: Product and Discount Impact

[Insert screenshot: Product & Discount Impact page]

Visuals: scatter plot of Discount % vs Profit (size = sales, red = loss, green = profit), product ranking table, and delivery time analysis by ship mode.

High-discount products harming profit Margin falls from 12.55% (5% to 10% discount) to 10.44% (15% to 20% discount). At product level the effect is weak, because every product's average discount is 9.5% to 10.3%. Tables - India is the only loss-making product (about -₹0.53 Cr, -8.5% margin) at a normal discount, so its loss comes from cost or pricing.

Delivery delays and trends Average delivery is 17.5 days (range 5 to 30) in every ship mode; Same Day (17.43) is no faster than Standard (17.51). The yearly average is flat (17.57, 17.52, 17.46). About 19% of orders take more than 25 days (the threshold used in this report).

Page 4: Region and Manager View

[Insert screenshot: Region & Manager View page]

Visuals: Region to State to City drill-down, profit by region manager, sales trend by region, bookmark buttons and navigation buttons.

Regional performance:

- North (Aniket Sharma): sales ₹33.21 Cr, profit ₹3.69 Cr, margin 11.1% - best by profit
- East (Yash Srivastava): sales ₹24.23 Cr, profit ₹2.87 Cr, margin 11.9%
- South (Harish Deshmukh): sales ₹24.07 Cr, profit ₹2.83 Cr, margin 11.8%
- West (Pragya Rathi): sales ₹13.81 Cr, profit ₹1.58 Cr, margin 11.4%
- Central (Nithin Chaturvedi): sales ₹9.61 Cr, profit ₹1.21 Cr, margin 12.6% - best margin

Which region and manager performed best? North and Aniket Sharma by profit (₹3.69 Cr); Central and Nithin Chaturvedi by margin (12.6%), although Central is the smallest region.

Worst performing territories States by margin: Uttar Pradesh (10.1%), Delhi (10.1%), Karnataka (10.4%) and Rajasthan (10.5%). Cities: Agra (6.2%) and Hubli (6.5%). No state or city has a net loss. North's profit fell from ₹1.36 Cr (2023) to ₹1.09 Cr (2025).

5. Phase 4: Advanced Power BI Features

Static Row-Level Security

Five roles were created, each filtering the Zone table by Region Manager. Because Zone is related to Orders, each filter restricts every page to one region. The roles were tested with View as in Power BI Desktop.

- North_Aniket_Sharma: [Region Manager] = "Aniket Sharma"
- East_Yash_Srivastava: [Region Manager] = "Yash Srivastava"
- South_Harish_Deshmukh: [Region Manager] = "Harish Deshmukh"
- West_Pragya_Rathi: [Region Manager] = "Pragya Rathi"
- Central_Nithin_Chaturvedi: [Region Manager] = "Nithin Chaturvedi"

[Insert screenshot: Manage roles window and View as test]

Bookmarks, tooltips, drill-through and navigation

- Bookmarks: Profit View and Sales View switch the region manager chart on Page 4.
- Custom tooltips: a customer tooltip page and a product tooltip page.
- Drill-through pages: Customer Profile and Region Details.
- Navigation buttons on every main page.

7. Key Business Recommendations

1. Grow Technology, which has the highest sales and the best margin (14.0%).
2. Review Furniture pricing, especially Tables, which lose about ₹0.53 Cr and pull Furniture margin down to 6.9%.
3. Plan stock and campaigns for August to December, which carry 57% of yearly sales, and run promotions in January and February.
4. Improve delivery performance: 19% of orders take over 25 days, and premium ship modes are no faster than Standard.
5. Recover margin in weak territories (Agra, Hubli, Uttar Pradesh, Karnataka) and reverse North's falling profit.
6. Reward high-profit customers and review discounts for loss-making ones: 84 customers are unprofitable, while customers such as Sana Bhattacharya earn the most profit.

8. Summary of Interactive Features

The following features are mandatory for this assignment and have all been implemented.

- **Static RLS** (whole report): five roles filter Zone[Region Manager], so each manager sees only their own region. Evidence: Manage roles and Test as role.
- **Bookmarks** (Region & Manager View): Profit View and Sales View buttons show or hide two charts to switch the view. Evidence: Bookmarks pane.
- **Drill-through** (Customer Insights to Customer Profile; Region & Manager View to Region Details): right-click a customer or region and drill through to a page filtered to it. Evidence: Customer Profile and Region Details pages.
- **Custom tooltips** (customer charts and product visuals): hovering shows a small report page with sales, profit and orders (customer) or sales, profit, profit % and discount % (product). Evidence: tooltip pages.
- **Navigation buttons** (all main pages): header buttons jump between the Sales, Customers, Products and Region pages. Evidence: header on each page.

Conclusion

The dashboard answers every expected insight in the assignment. Technology leads on sales and margin; sales are evenly spread across states but seasonal and flat overall; North is the strongest region and Central the most efficient; and delivery performance and a few low-margin territories are the main areas for improvement. Row-Level Security ensures each Regional Manager sees only their own region.